Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Inequalities

Name _____ Date _____ No calculator Version 1

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KEY WORDS inequality *(whārite-kore)* · greater than *(rahi ake i)* · less than *(iti iho i)* · number line *(rārangi tau)*

A Writing inequalities ●○○

Remember: $<$ less than, $>$ greater than, $\leq$ less than or equal to, $\geq$ greater than or equal to.

1. Write each sentence as an inequality.

a) x is greater than 4 _____

b) x is less than 9 _____

c) x is at least 2 _____

d) x is at most 10 _____

e) x is more than −1 _____

f) x is no more than 6 _____

g) x is no less than −5 _____

h) x is under 0 _____

2. Write an inequality for each situation. Use the letter given.

a) You must be at least 12 years old to go on a ride. (age a) _____

b) A lift can carry at most 8 people. (number of people n) _____

c) The temperature stayed below 5°C all day. (temperature t) _____

d) A bag must weigh more than 2 kg to need a label. (mass m) _____

B Which numbers fit? ●○○

3. Circle every number that makes the inequality true.

a) $x > 5$: 3, 5, 5.5, 9

b) $x \leq 2$: -1, 0, 2, 3

c) $x \geq -3$: -4, -3, -2.5, 0

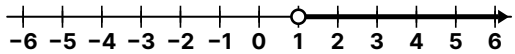
d) $x < 0$: -2, -0.5, 0, 1

C Number lines ●●○

Remember: An open circle means the number is not included ($<$ or $>$). A filled circle means it is included ($\leq$ or $\geq$).

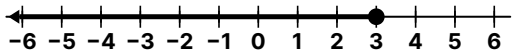
4. Write the inequality shown on each number line.

a)



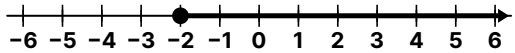
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b)



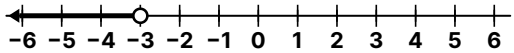
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c)



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d)

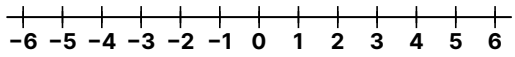


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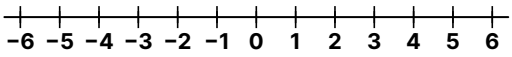
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5. Show each inequality on its number line.

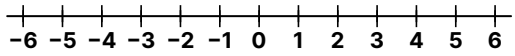
a) $x > -2$



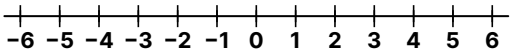
b) $x \leq 5$



c) $x \geq 0$



d) $x < 4$



D Solving one-step inequalities ●●○

Remember: Solve an inequality the same way as an equation: do the same to both sides, and keep the inequality sign in every line.

6. Solve.

a) $x + 4 > 9$ **b)** $x - 3 \leq 3$ **c)** $3x < 12$ **d)** $5x \geq 15$

e) $\frac{x}{2} > 5$ **f)** $x + 7 < 5$ **g)** $4x \leq -12$ **h)** $\frac{x}{3} \geq 4$

E Solving two-step inequalities ●●●

7. Solve. Show each step.

a) $2x + 1 > 9$ **b)** $3x - 2 \leq 13$ **c)** $4x + 5 < 13$ **d)** $2x - 7 \geq 5$

e) $5x + 3 > -2$ **f)** $6x - 5 < -5$ **g)** $\frac{x}{4} + 1 \geq 3$ **h)** $\frac{x}{2} - 3 \leq 2$

★ Extension

8. The solution of an inequality is $x > 2$. Which of the whole numbers 0, 1, 2, 3, 4 and 5 are solutions?

9. Aroha has $40. She pays $4 to get into a fair, and each ride costs $6. Write an inequality for the number of rides x she can go on, and solve it. What is the largest number of rides?

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Inequalities

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KEY WORDS inequality *(whārite-kore)* · greater than *(rahi ake i)* · less than *(iti iho i)* · number line *(rārangi tau)*

A Writing inequalities ●○○

Remember: $<$ less than, $>$ greater than, $\leq$ less than or equal to, $\geq$ greater than or equal to.

1. Write each sentence as an inequality.

a) x is greater than 6 _____

b) x is less than 7 _____

c) x is at least 3 _____

d) x is at most 12 _____

e) x is more than −2 _____

f) x is no more than 8 _____

g) x is no less than −4 _____

h) x is under 0 _____

2. Write an inequality for each situation. Use the letter given.

a) You must be at least 10 years old to join a club. (age a) _____

b) A van can carry at most 6 people. (number of people n) _____

c) The temperature stayed below 3°C all night. (temperature t) _____

d) A parcel must weigh more than 5 kg to cost extra. (mass m) _____

B Which numbers fit? ●○○

3. Circle every number that makes the inequality true.

a) $x > 4$: 2, 4, 4.5, 7

b) $x \leq 3$: -2, 0, 3, 4

c) $x \geq -2$: -3, -2, -1.5, 0

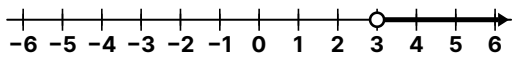
d) $x < 1$: -3, 0.5, 1, 2

C Number lines ●●○

Remember: An open circle means the number is not included ($<$ or $>$). A filled circle means it is included ($\leq$ or $\geq$).

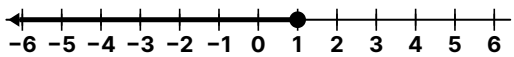
4. Write the inequality shown on each number line.

a)



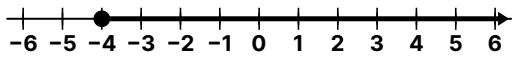
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b)



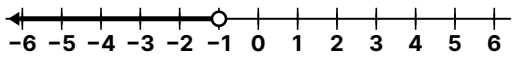
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c)



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d)

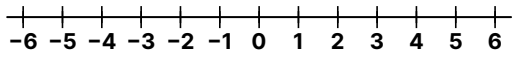


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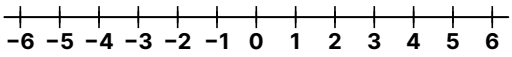
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5. Show each inequality on its number line.

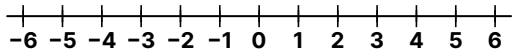
a) $x > -4$



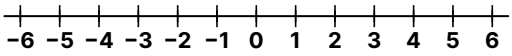
b) $x \leq 3$



c) $x \geq 1$



d) $x < 2$



D Solving one-step inequalities ●●○

Remember: Solve an inequality the same way as an equation: do the same to both sides, and keep the inequality sign in every line.

6. Solve.

a) $x + 6 > 9$

b) $x - 5 \leq -1$

c) $4x < 12$

d) $2x \geq 14$

e) $\frac{x}{3} > 3$

f) $x + 9 < 6$

g) $5x \leq -10$

h) $\frac{x}{2} \geq 7$

E Solving two-step inequalities ●●●

7. Solve. Show each step.

a) $3x + 1 > 7$

b) $2x - 5 \leq 7$

c) $5x + 3 < 13$

d) $3x - 8 \geq 4$

e) $4x + 3 > -1$

f) $7x - 6 < -6$

g) $\frac{x}{5} + 2 \geq 4$

h) $\frac{x}{3} - 1 \leq 3$

★ Extension

8. The solution of an inequality is $x > 3$. Which of the whole numbers 0, 1, 2, 3, 4 and 5 are solutions?

9. Tane has $45. He pays $5 to get into a fair, and each ride costs $8. Write an inequality for the number of rides x he can go on, and solve it. What is the largest number of rides?

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